

Carola Bonamico

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Education

- Politecnico di Torino**, MSc in Computer Engineering – Software Sep 2024 – Present
- Weighted average: 27.8/30
 - Main courses: Data Science and Database Technology, Software Engineering I/II, Web Applications I, Large Language Models for Software Engineering, Human Computer Interaction, Computer Architectures, Information Systems, System and Device Programming
- University of Pavia**, BSc in Biomedical Engineering Sep 2020 – Apr 2024
- Thesis: “Literature review on the impact of sleep quality on glycemic patterns in patients with type 1 diabetes”
 - Main courses: Biomedical Data Processing, Models of Biological Systems
- I.I.S. Cellini – Liceo Linguistico Alberti**, Secondary School Diploma in Modern Languages Sep 2015 – Jun 2020
- Languages studied: English, French, German

Experience

- Master’s Thesis Student**, ETH Zurich – Zurich, CH Mar 2026 – Sep 2026
- Conducted Master’s thesis project at the Integrated Systems Laboratory (IIS), focused on developing edge-constrained Deep Learning models for silent speech recognition from electromyographic signals.
 - Reached 85.8% and 82.1% accuracy (vocalised, silent) in closed-set classification and 16.3% and 22.4% Word Error Rate (WER) in continuous recognition.
 - Designed and collected a multimodal, multi-session dataset: 7 subjects, 6 sessions each, about 40 hours of synchronised EMG, EEG and audio covering 15 words and 20 sentences in both vocalised and silent conditions.
- Software Developer**, Squadra Corse PoliTo – Turin, IT Aug 2025 – Nov 2025
- Built a TUI in Go to send, receive and inspect CAN messages over the SocketCAN interface, streamlining on-track diagnostics and debugging.
- Academic Tutor**, University of Pavia – Pavia, IT Sep 2022 – Oct 2023
- Tutored small groups of students on the Fundamentals of Computer Science course and gave targeted one-to-one support on C programming exercises, covering algorithm design and memory management.

Technical Skills

Programming languages: C, Rust, Go, JavaScript, TypeScript, Python, ARM Assembly, MATLAB, ~~TeX~~

Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn

Libraries and frameworks: Node.js, React.js, React Native, TypeORM, Diesel

Databases: Oracle SQL, MongoDB, MySQL, SQLite, PostgreSQL

Tools and software: Git, GitHub, GitLab, Docker, Jupyter Notebook, LabVIEW

Other: REST APIs, WebSocket, JSON, Repository Pattern, Agile, UML, HTML, CSS

Personal Projects

-  **CAN debug TUI**
CAN bus debugging TUI written in Go, sending and receiving CAN messages over the SocketCAN interface with per-message control. Supports DBC files for message decoding.
-  **Autostar Motorsport Web App**
Responsive web application for a motorsport company, built with Node.js and React.js. Implements a client-side architecture in JavaScript, GDPR compliance through Iubenda and modular React components styled with CSS and AOS.

Projects at ETH Zurich

-  **SilentWear**
Fork of SilentWear, a wearable system based on a textile neckband for silent speech recognition from EMG. Extended the existing CNN with a BiLSTM and a Transformer, added CTC decoding for direct EMG-to-text recognition alongside closed-set classification, and implemented an EMG-based automatic speech onset detection algorithm.
-  **BioGUI**
Contributor to BioGUI, an open source platform for real-time biosignal acquisition and labelling. Extended the data collection pipeline to support a custom vocabulary protocol. Implemented intra-file alignment across EMG, EEG and microphone signals in the binary .bio format, as well as inter-file alignment across multiple .bio recordings.

Projects at Politecnico di Torino

-  **Participium**
Full-stack platform for civic issue reporting in TypeScript/JavaScript (Node.js and web frontend), with role-based flows, geolocation and photo upload. Includes Telegram notifications and SonarQube for static analysis and continuous code quality monitoring.

Rust Chat

Real-time group messaging application built with Rust and React. Supports private chats, groups, JWT authentication, WebSocket communication and Docker deployment.

HCI MoveMate

Repository of documentation and reports on the HCI research and UX design process, including findings, decision rationale and evaluation outcomes supporting the MoveMate prototype.

MoveMate

High-fidelity mobile app prototype built with React Native (Expo), focused on discovering and joining community events through an end-to-end mobile UX flow.

Geocontrol

Dockerised software monitoring system for managing geological, meteorological and environmental sensors. Built in TypeScript using the repository pattern, TypeORM and Node.js.

Class Assignments Manager

Web application for managing assignments given by teachers to groups of students. Server side in JavaScript; client side in React.js, HTML and React Bootstrap with CSS.

Pac-Man on LandTiger

Pac-Man game for the LandTiger LPC1768 board (ARM Cortex-M3), with an A* algorithm for ghost movement, a pause button and in-game audio through a speaker. Written in C and ARM assembly.

Projects at University of Pavia

Clinical Trial Manager

Website for managing clinical trials in HTML and CSS, supporting communication between doctors, patients and clinical managers, and trial monitoring and follow-up.

Pulse Oximeter Data Analysis

LabVIEW software for processing data acquired from a pulse oximeter designed and assembled in the laboratory.

Healthcare Management Application

MATLAB application for managing the activities of regions, local health authorities and hospitals, with statistical analysis of patients' glycemic curves through XML file processing.

Certifications

Goethe-Zertifikat B1 (German), Goethe-Institut Italia

Issued: Jul 2019

B1 Preliminary (English), Cambridge English

Issued: May 2018

Delf A2 (French), Ministère de l'Éducation nationale

Issued: Sep 2015